SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I T NITHIN KUMAR(192424351) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: NITHIN KUMAR

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

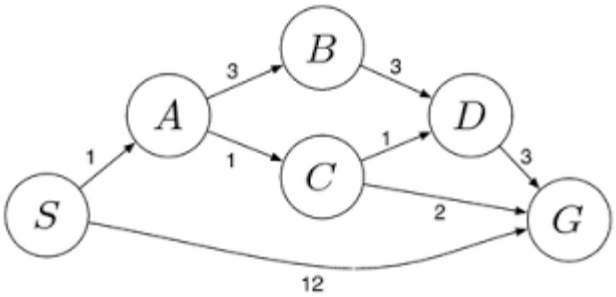
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

		accurately			

1. Water Jug Problem

1.1 Problem Statement

You are given 2 jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug?

1.2 Assumptions

- A jug can be filled from the pump.
- Water can be poured out of a jug onto the ground.
- Water can be poured from one jug to another.
- There are no other measuring devices available.

(These are the explicit assumptions given in the question paper. No other assumption is made.)

1.3 State Representation

A state is written as an ordered pair **(x, y)**, where:

- x = gallons of water in the 4-gallon jug ($0 \le x \le 4$)
- y = gallons of water in the 3-gallon jug ($0 \le y \le 3$)
- **Initial state:** (0, 0) – both jugs are empty.
- **Goal state:** (2, y) – exactly 2 gallons in the 4-gallon jug (any y).

1.4 Operators (Production Rules)

No.	Rule	Action
1	$(x, y), x < 4 \rightarrow (4, y)$	Fill the 4-gallon jug from the pump
2	$(x, y), y < 3 \rightarrow (x, 3)$	Fill the 3-gallon jug from the pump
3	$(x, y), x > 0 \rightarrow (0, y)$	Empty the 4-gallon jug on the ground
4	$(x, y), y > 0 \rightarrow (x, 0)$	Empty the 3-gallon jug on the ground
5	$(x, y), x + y \ge 4, y > 0 \rightarrow (4, y - (4 - x))$	Pour from 3-gallon jug into 4-gallon jug until it is full
6	$(x, y), x + y \ge 3, x > 0 \rightarrow (x - (3 - y), 3)$	Pour from 4-gallon jug into 3-gallon jug until it is full
7	$(x, y), x + y \le 4, y > 0 \rightarrow (x + y, 0)$	Pour all water from 3-gallon jug into 4-gallon jug
8	$(x, y), x + y \le 3, x > 0 \rightarrow (0, x + y)$	Pour all water from 4-gallon jug into 3-gallon jug

1.5 Sequence of States and Operations

Step	State (x, y)	Operation	Rule Used
0	(0, 0)	Initial state – both jugs empty	–
1	(0, 3)	Fill the 3-gallon jug from the pump	Rule 2
2	(3, 0)	Pour all 3 gallons from the 3-gallon jug into the 4-gallon jug	Rule 7
3	(3, 3)	Fill the 3-gallon jug again from the pump	Rule 2
4	(4, 2)	Pour from the 3-gallon jug into the 4-gallon jug until it is full (only 1 gallon fits, so 2 gallons stay in the 3-gallon jug)	Rule 5
5	(0, 2)	Empty the 4-gallon jug on the ground	Rule 3
6	(2, 0)	Pour all 2 gallons from the 3-gallon jug into the 4-gallon jug	Rule 7

1.6 How Exactly 2 Gallons Are Obtained in the 4-Gallon Jug

- After Step 3, the 4-gallon jug holds 3 gallons, so it has room for only 1 more gallon ($4 - 3 = 1$).
- In Step 4, only 1 gallon moves out of the full 3-gallon jug. The remaining water in it is $3 - 1 = 2$ gallons.
- Step 5 empties the 4-gallon jug, so the 2 gallons are now alone in the 3-gallon jug.
- Step 6 pours these 2 gallons into the empty 4-gallon jug, giving exactly 2 gallons in it.

Calculation: $3 + 3 = 6$; $6 - 4 = 2$ (total water drawn from the pump minus the 4 gallons discarded).

1.7 Final State

(2, 0) – the 4-gallon jug contains exactly 2 gallons and the 3-gallon jug is empty. The goal is reached in 6 operations.

1.8 Conclusion

By representing the problem as states (x, y) and applying the permitted fill, empty and pour operations, exactly 2 gallons were measured in the 4-gallon jug without any marker on either jug.

2. Mars Rover – Intelligent Agent

The question asks how a Mars Rover analyses and explores samples on Mars. The Mars Rover is treated as an intelligent agent that senses its surroundings through sensors and acts on them through actuators.

2.1 (i) Percepts of the Agent

Percepts are the inputs the rover receives from its environment through its sensors:

- Camera images of the terrain, rocks, soil and obstacles.
- Distance and obstacle information (for example, from range or proximity sensors).
- Slope, tilt and wheel-contact readings of the ground.
- Position, direction and speed of the rover (navigation sensors).
- Chemical and mineral readings of rock and soil samples from its science instruments.
- Temperature, dust and light conditions on Mars.
- Battery level and the health or status of rover parts.
- Commands and messages received from the Earth control station.

2.2 (ii) Characterisation of the Operating Environment

Property	Type	Reason
Observability	Partially observable	Sensors see only the nearby area; the full map of Mars is not known.
Agents	Single agent	Only the rover acts; no other agent competes with it.
Determinism	Stochastic	Wheel slip on sand, dust and rough ground make outcomes uncertain.
Episodic / Sequential	Sequential	Present moves affect later moves (position, battery, wear).
Static / Dynamic	Dynamic	Dust storms, temperature and lighting change while the rover acts.
Discrete / Continuous	Continuous	Positions, speeds and sensor values take continuous values.
Known / Unknown	Largely unknown	Terrain and sample content are discovered during exploration.

2.3 (iii) Actions the Agent Can Take

- Move forward, move backward, turn left, turn right, and stop.
- Change speed or choose a different path to avoid obstacles and steep slopes.
- Take pictures and point the camera or mast.

- Extend the robotic arm and drill, scoop or collect rock and soil samples.
- Analyse samples with onboard instruments.
- Store samples and record results.
- Send data and images to Earth, and receive commands.
- Charge the battery (solar panels) and go into a low-power or safe mode.

2.4 (iv) Evaluating the Performance of the Agent

The performance measure judges how well the rover achieves its mission. It can be evaluated using:

Criterion	How it is measured
Samples analysed	Number and quality of useful samples collected and analysed
Area explored	Distance covered and area of terrain surveyed
Safety	No collisions, falls into craters or getting stuck
Energy efficiency	Battery and power used per task completed
Data returned	Amount of correct data and images sent back to Earth
Time and reliability	Time taken to complete tasks and how long the rover keeps working

2.5 (v) Most Suitable Agent Architecture and Why

Suitable architecture: Goal-based agent combined with a utility-based element, using a model of the world (a model-based agent).

- A simple reflex agent is not enough, because the environment is only partially observable and the rover must remember its map, position and past readings (model-based).
- The rover has clear goals such as reaching a target rock or exploring an area, so it must plan sequences of moves (goal-based).
- When several options exist, it must choose the best one, for example the safest and lowest-energy path, or the most valuable sample (utility-based).
- Communication delay with Earth means the rover must decide many actions by itself, so it needs this reasoning ability.
- It should also be able to learn from experience, for example about difficult terrain.

3. 8-Queens Problem

3.1 Explanation of the Problem

Place 8 queens on a chessboard (8 × 8) such that none of the queens attacks any of the others. A queen attacks any other queen in the same row, same column or same diagonal. The question states that a configuration of 8 queens is shown in a figure but is not a solution, because the queen in the first column is on the same diagonal as the queen in the last column.

Note: The figure referred to in the question paper ("the figure below") is not present in the uploaded document, so that particular configuration cannot be reproduced or analysed here. Only the reason given in the question text is used.

3.2 Problem Formulation

Using the incremental formulation (queens are added one at a time, one per column):

Component	Definition
State space	Any arrangement of 0 to 8 queens on the board. With the improved formulation, a state has n queens placed in the leftmost n columns, with no two queens attacking each other.
Initial state	The empty board – no queens placed.
Actions / Operators	Add a queen to a square in the leftmost empty column such that it is not attacked by any queen already on the board.
Transition model	Returns the board with the new queen added in the chosen square.
Goal test	8 queens are on the board and none is attacked.
Path cost	Each action (placing a queen) has cost 1. Only the final arrangement matters, so path cost is not important here. A complete solution always has path cost 8.

3.3 Constraints

Let a queen be at row r and column c.

- **Row constraint:** no two queens can have the same row number r.
- **Column constraint:** no two queens can have the same column number c.
- **Diagonal constraint:** two queens at (r1, c1) and (r2, c2) are on the same diagonal if $|r1 - r2| = |c1 - c2|$; equivalently, $r - c$ is the same (one direction) or $r + c$ is the same (other direction). All queens must have different $r - c$ values and different $r + c$ values.

3.4 How the Search Is Performed

Uninformed search (depth-first search with backtracking) can be used:

1. Start with the empty board.
2. Place a queen in the first column in the first safe row.
3. Move to the next column and place a queen in the first row that is not attacked.

4. If a column has no safe row, backtrack to the previous column and move that queen to its next safe row.
5. Continue until 8 queens are placed (goal test satisfied), or all possibilities are exhausted.

Checking each queen against the earlier ones prunes unsafe branches early, which makes the search far smaller than trying all possible placements of 8 queens.

3.5 One Valid 8-Queens Solution

Queen positions (row, column): (1,1), (2,5), (3,8), (4,6), (5,3), (6,7), (7,2), (8,4)

	1	2	3	4	5	6	7	8
1	Q							
2					Q			
3								Q
4						Q		
5			Q					
6							Q	
7		Q						
8				Q				

Rows are numbered 1–8 (left) and columns 1–8 (top).

3.6 Why the Solution Is Valid

Queen	Row r	Column c	r − c	r + c
1	1	1	0	2
2	2	5	−3	7
3	3	8	−5	11
4	4	6	−2	10
5	5	3	2	8
6	6	7	−1	13
7	7	2	5	9
8	8	4	4	12

- **Rows:** the rows 1 to 8 are each used exactly once, so no two queens share a row.
- **Columns:** the columns 1, 5, 8, 6, 3, 7, 2, 4 are all different (each of 1–8 appears once), so no two queens share a column.

- **Diagonals (r − c):** 0, −3, −5, −2, 2, −1, 5, 4 are all different, so no two queens share a "\" diagonal.
- **Diagonals (r + c):** 2, 7, 11, 10, 8, 13, 9, 12 are all different, so no two queens share a "/" diagonal.

Check against the concern in the question: the queen in the first column is at (1,1) and the queen in the last column is at (3,8). Here $|1 - 3| = 2$ but $|1 - 8| = 7$, so they are not on the same diagonal. Every other pair is also checked by the distinct r − c and r + c values above.

3.7 Conclusion

The 8-Queens problem was formulated with a state space, initial state, actions, transition model, goal test and path cost. Depth-first search with backtracking can find solutions, and the arrangement given above satisfies all row, column and diagonal constraints.

4. OLA Cab Booking

4.1 Problem Statement

A customer wants to travel from one location to another using the OLA Cab booking mobile application. The customer can select any cab type such as mini, micro, sedan, shared, prime and so on, based on his comfort, to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

4.2 Most Suitable Agent

A goal-based problem-solving agent with a utility-based selection of the cab type. It is best described as a goal-based, utility-based agent.

4.3 Why This Agent Is Suitable

- The agent has a clear goal: take the customer from the source to the destination.
- The agent must choose among several alternatives (mini, micro, sedan, shared, prime, etc.) instead of reacting with a fixed rule, so a simple reflex agent is not suitable.
- Each cab type differs in fare, travel time, comfort and availability, so the options must be compared and the one that best fits the customer's preference must be chosen. This comparison is what a utility-based agent does.
- The environment (cab availability, traffic, fares) can change, so the agent must check the current situation before selecting.

4.4 PEAS Description

Element	Description
Performance measure	Customer reaches destination with low fare, short travel time and good comfort; booking is successful

Element	Description
Environment	OLA app, available drivers and cabs, roads and traffic, customer
Actuators	App screen (shows options), booking confirmation, driver assignment request
Sensors	Source and destination input, cab type choice, location, availability and fare data

4.5 Factors Considered

Factor	Use in decision
Source and destination	Define the start and goal of the trip
Cab types	Mini, Micro, Sedan, Shared, Prime and so on
Fare	Estimated cost of each cab type for the trip
Travel time	Estimated time to reach the destination
Comfort	Level of comfort offered by the cab type
Availability	Cab must be available near the source
Customer preference	Customer's priority (low fare, fast, or comfort)

No numerical fares, times or weights are given in the question paper, so none are assumed. The agent reads these values from the application when booking.

4.6 Pseudocode

```
ALGORITHM OlaCabBooking
INPUT : source, destination, customer_preference
        (preference = LOW_FARE / FAST / COMFORT)
OUTPUT: booked cab type with fare and travel time, or message

1.  START
2.  READ source and destination
3.  IF source or destination is invalid THEN
        DISPLAY "Enter valid locations"; STOP
4.  cab_list <- {Mini, Micro, Sedan, Shared, Prime, ...}
5.  available_list <- empty
6.  FOR each cab_type IN cab_list DO
        IF cab_type is available near source THEN
            fare <- EstimateFare(cab_type, source, destination)
            time <- EstimateTime(cab_type, source, destination)
            comfort <- ComfortLevel(cab_type)
            ADD (cab_type, fare, time, comfort) TO available_list
        END IF
    END FOR
7.  IF available_list is empty THEN
        DISPLAY "No cabs available"; STOP
8.  DISPLAY available_list to customer
9.  READ customer_preference
10. FOR each option IN available_list DO
        option.score <- Utility(option, customer_preference)
    END FOR
```

```
11. best <- option with the highest score
12. DISPLAY best and ASK customer to confirm or choose another cab
13. IF customer confirms THEN
        BOOK best cab; ASSIGN driver
        DISPLAY "Booking confirmed", cab type, fare, time
    ELSE
        GO TO step 8
    END IF
14. STOP
```

4.7 Input, Processing and Output

Stage	Details
Input	Source, destination, customer preference, available cab types, and their fare, time and comfort values
Processing	Check availability; estimate fare and time for each cab; compare options with the customer's preference (utility); select the best option; take confirmation
Output	Confirmed booking with cab type, estimated fare, travel time and driver details (or "no cabs available")

4.8 Final Result

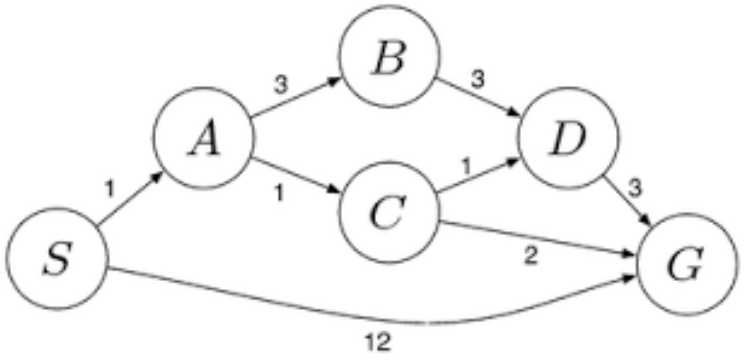
A goal-based, utility-based problem-solving agent can book the most suitable OLA cab by checking availability, comparing fare, travel time and comfort, and choosing the cab type that best matches the customer's preference.

5. Uniform Cost Search

5.1 Problem Statement

A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

5.2 The Weighted Graph (as given in the question paper)



Nodes: S, A, B, C, D, G. **Start:** S. **Goal:** G.

Edges are directed (arrows in the figure) and the costs are exactly as shown:

From	To	Cost
S	A	1
S	G	12
A	B	3
A	C	1
B	D	3
C	D	1
C	G	2
D	G	3

5.3 Uniform Cost Search Algorithm

UCS is an uninformed search that always expands the node in the frontier with the lowest cumulative path cost g(n).

1. Put the start node S in the frontier (priority queue) with cost 0.
2. Remove the node with the lowest cumulative cost from the frontier.
3. If this node is the goal G, stop; the path found is the least-cost path.
4. Otherwise, mark it as expanded and generate its successors. For each successor, cost = cost of current node + edge cost.
5. Add successors to the frontier. If a node is already in the frontier with a higher cost, replace it by the cheaper one. Ignore nodes already expanded.

6. Repeat from step 2.

Note: the goal test is applied when a node is removed (expanded), not when it is generated. If two nodes have equal cost, the one inserted first is expanded first.

5.4 Step-by-Step Execution

Step	Node expanded (g)	Successors generated (cumulative cost)	Frontier after step (node : cost)	Expanded list
1	S (0)	A = 0 + 1 = 1; G = 0 + 12 = 12	A:1, G:12	S
2	A (1)	B = 1 + 3 = 4; C = 1 + 1 = 2	C:2, B:4, G:12	S, A
3	C (2)	D = 2 + 1 = 3; G = 2 + 2 = 4 (better than 12, replaces it)	D:3, B:4, G:4	S, A, C
4	D (3)	G = 3 + 3 = 6 (worse than 4, ignored)	B:4, G:4	S, A, C, D
5	B (4)	D = 4 + 3 = 7 (D already expanded, ignored)	G:4	S, A, C, D, B
6	G (4)	Goal reached – stop	–	S, A, C, D, B, G

Order of expansion: S → A → C → D → B → G

5.5 Cumulative Path Costs and Comparison of Paths

No.	Path from S to G	Cost calculation	Total cost
1	S → A → C → G	1 + 1 + 2	4
2	S → A → C → D → G	1 + 1 + 1 + 3	6
3	S → A → B → D → G	1 + 3 + 3 + 3	10
4	S → G	12	12

Path 1 has the lowest total cost. UCS finds it without listing all paths, because it always expands the cheapest node first and G is removed from the frontier with cost 4 before any costlier route can be completed.

5.6 Least-Cost Path and Final Minimum Cost

Least-cost path: S → A → C → G

Minimum cost = 1 + 1 + 2 = 4

5.7 Conclusion

Uniform Cost Search expanded the nodes in order S, A, C, D, B, G and found that the least-cost route for the delivery network from warehouse S to warehouse G is S → A → C → G with a total cost of 4. The direct route S → G (cost 12) is much more expensive, which shows that the fewest edges does not always mean the lowest cost.

6. Final Section

6.1 Overall Result

Q.No.	Topic	Result
1	Water Jug Problem	Final state (2, 0): 2 gallons in the 4-gallon jug after 6 operations
2	Mars Rover Agent	Percepts, environment, actions and performance measures identified; goal-based, model-based, utility-based architecture suggested
3	8-Queens Problem	Problem formulated; valid solution (1,1) (2,5) (3,8) (4,6) (5,3) (6,7) (7,2) (8,4)
4	OLA Cab Booking	Goal-based, utility-based agent; pseudocode written
5	Uniform Cost Search	Least-cost path S → A → C → G with minimum cost 4

6.2 Key AI Concepts Used

- State space representation and production rules (Q1)
- Intelligent agent, percepts, actions, PEAS and environment properties (Q2)
- Agent types: reflex, model-based, goal-based and utility-based (Q2, Q4)
- Problem formulation – initial state, actions, transition model, goal test, path cost (Q3)
- Uninformed search: depth-first search with backtracking (Q3) and Uniform Cost Search (Q5)
- Priority queue (frontier) and cumulative path cost (Q5)

6.3 Short Conclusion

All five questions were solved using the basic AI concepts of CO1. Problems were represented as states and actions, intelligent agents were analysed and selected, and uninformed search methods were applied step by step to obtain correct solutions.

Signature: _____ Date: _____